

ALIZA BEVERAGE

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RESEARCH INTERESTS

The formation and evolution of galaxies, stars and stellar populations, star-formation quenching, origins of the elements, chemical evolution

PROFESSIONAL APPOINTMENTS

Hubble Fellow, Carnegie Observatories	September 2025–present
Carnegie-Princeton Fellow, Carnegie Observatories	September 2025–present

EDUCATION

University of California Berkeley, <i>Ph.D, Astrophysics</i>	August 2025
<i>Advisors: Mariska Kriek and Dan Weisz</i>	
University of California Berkeley, <i>Masters of Arts, Astrophysics</i>	August 2020
University of Minnesota Twin Cities, <i>Bachelor of Science, Physics and Astrophysics</i>	June 2018
<i>Advisor: Claudia Scarlata</i>	

SELECTED AWARDS AND HONORS

NSF Graduate Research Fellowship	2019–2024
Robert J. Trumpler Graduate Student Excellence Award	2024
<i>In recognition of academic excellence and outstanding record of involvement in the Berkeley department or wider astronomical community.</i>	
H2H8 Fellowship (\$10,000)	2022
AAS International Travel Grant (\$6,000)	2021, 2023, 2024
Outstanding Astronomy Graduate Student Instructor	2021
Maria Cranor Fellowship (\$18,000)	2019–2025
Astronaut Scholarship (\$10,000)	2018–2019

OBSERVING PROGRAMS AS PRINCIPAL INVESTIGATOR

<i>JWST</i> , NIRSpec (GO 11451 ; \$310,000)	43.9 hrs, Cycle 5
<i>How Fast Is Too Furious? Ultradeep Spectroscopy of a Fossil from Cosmic Dawn</i>	
<i>JWST</i> , NIRCам (co-PI with Chloe Cheng; GO 11532 ; \$72,000)	5.4 hrs, Cycle 5
<i>It's the Little Things that Count: Unveiling Faint Merger Signatures and Tiny Companions around Massive Quiescent Galaxies at $z \sim 2$</i>	
<i>JWST</i> , NIRSpec (co-PI with Mariska Kriek and Chloe Cheng; GO 5629)	40.3 hrs, Cycle 3
<i>Extremely Deep Spectroscopy of Quiescent Galaxies at $z \sim 0.7$: A Direct Measurement of the Stellar Initial Mass Function Beyond the Low-Redshift Universe</i>	
<i>JWST</i> , NIRSpec (co-PI with Mariska Kriek; GO 2110)	22.7 hrs, Cycle I
<i>Ultra-Deep Continuum Spectroscopy of Quiescent Galaxies at $1.0 < z < 2.5$: Chemical Abundances and Stellar Kinematics</i>	
Magellan, LLAMAS	9 nights, 2025B - 2026B
<i>The Anemic Survey: Spatially Resolving the Chemical Abundances, Kinematics, and IMF of Quiescent Galaxies</i>	

OBSERVING PROGRAMS AS CO-INVESTIGATOR

<i>JWST</i> , NIRSpec and NIRCам (PI: Katherine Whitaker; GO 11883) <i>Opening a New Window on Globular Cluster Formation with Strong Lensing at $z = 2$</i>	32.9 hrs, Cycle 5
<i>JWST</i> , NIRSpec (PI: Elham Eftekhari; GO 10660) <i>Unlocking the Power of the Near-Infrared: Detailed Measurement of the Stellar Initial Mass Function in Quiescent Galaxies at Intermediate-z</i>	38.2 hrs, Cycle 5
<i>JWST</i> , NIRSpec and NIRCам (PI: Nathan Sandford; GO 10246) <i>Exploring the Accretion History and Halo of Giant Elliptical Cent A with Resolved Stars</i>	131.3 hrs, Cycle 5
<i>JWST</i> , NIRSpec (PI: Mariska Kriek; GO 8317) <i>The Return of the Giants: Constraining the TP-AGB Phase across Cosmic Time</i>	7.9 hrs, Cycle 4
Keck, LRIS and MOSFIRE (PI: Mariska Kriek) <i>The Heavy Metal Survey: The chemical enrichment, star-formation and assembly histories of $z \sim 1.4 - 2.3$ quiescent galaxies</i>	10 nights 2020A-2021B

OPEN SOURCE SOFTWARE

alf α : Chemical abundance fitting code for old stellar systems ([A. Beverage 2024](#))
*A python adaption of **alf** (Conroy et al. 2018)*

PUBLICATION LIST

5 first-author, 6 second- and third-author papers ([ADS library](#))

First-author:

5. *Are Nucleosynthetic Yields Universal? Interpreting the Multi-Elemental Abundances of Quiescent Galaxies over Cosmic Time Using Milky Way Stars*
Beverage, A. G., D. Weinberg, M. Kriek, N. Marcelina Gountanis, A. Newman, D. Wiesz (Under review in ApJ)
4. *Carbon and Iron Deficiencies in Quiescent Galaxies at $z=1-3$ from JWST-SUSPENSE: Implications for the Formation Histories of Massive Galaxies*
Beverage, A. G., M. Slob, M. Kriek, C. Conroy, G. Barro, R. Bezanson, G. Brammer, C. M. Cheng, A. de Graaff, N. M. Förster Schreiber, M. Franx, B. Lorenz, P. E. Mancera Piña, D. Marchesini, A. Muzzin, A. B. Newman, S. H. Price, A. E. Shapley, M. Stefanon, K. A. Suess, P. van Dokkum, D. Weinberg, and D. R. Weisz, 2024, ApJ, 979, 2, DOI: [10.3847/1538-4357/ad96b6](#)
3. *The Heavy Metal Survey: The Evolution of Stellar Metallicities, Abundance Ratios, and Ages of Massive Quiescent Galaxies since $z \sim 2$*
Beverage, A. G., M. Kriek, K. A. Suess, C. Conroy, S. H. Price, G. Barro, R. Bezanson, M. Franx, B. Lorenz, Y. Ma, L. A. Mowla, I. Pasha, P. van Dokkum, and D. R. Weisz, 2024, ApJ, 966, 234, DOI: [10.3847/1538-4357/ad372d](#)
2. *From Carbon to Cobalt: Chemical Compositions and Ages of $z \sim 0.7$ Quiescent Galaxies*
Beverage, A. G., M. Kriek, C. Conroy, N. R. Sandford, R. Bezanson, M. Franx, A. van der Wel, and D. R. Weisz, 2023, ApJ, 948, 140, DOI: [10.3847/1538-4357/acc176](#)
1. *Elemental Abundances and Ages of $z \sim 0.7$ Quiescent Galaxies on the Mass-Size Plane: Implication for Chemical Enrichment and Star Formation Quenching*
Beverage, A. G., M. Kriek, C. Conroy, R. Bezanson, M. Franx, and A. van der Wel, 2021, ApJ, 917 L1, DOI: [10.3847/2041-8213/ac12cd](#)

Second- and Third-author:

6. *A Spectral Atlas for Quiescent Galaxies*
Hamadouche, M. L., J. Antwi-Danso, **A. Beverage**, and K. E. Whitaker, 2026, Res. Notes AAS, 10, 133, DOI: [10.3847/2515-5172/ae7077](https://doi.org/10.3847/2515-5172/ae7077)
5. *Cold Gas in a Post-Starburst Pair at $z \sim 1.4$: Major Mergers as a Pathway to Quenching in the HeavyMetal Survey*
Suess, K. A., **A. Beverage**, M. Kriek, J. S. Spilker, R. Bezanson, V. R. D’Onofrio, J. E. Greene, J. Lin, Y. Luo, D. Narayanan, I. Pasha, S. H. Price, D. J. Setton, M. E. Verrico, and Y. Zhang, submitted to ApJL, ([arXiv](https://arxiv.org/abs/2508.10001))
4. *Modeling the Ages and Chemical Abundances of Elliptical Galaxies*
Marcelina Gountanis, N., D. H. Weinberg, **A. Beverage**, N. R. Sandford, C. Conroy, and M. Kriek, submitted to ApJ, ([arXiv](https://arxiv.org/abs/2508.10001))
3. *Age and metal gradients in massive quiescent galaxies at $0.6 < z < 1.0$: Implications for quenching and assembly histories*
Cheng, C. M., Kriek, M., **A. Beverage**, van der Wel, A., Bezanson, R., D’Eugenio, F., Franx, M., Mancera Piña, P. E., Nersesian, A., Slob, M., Suess, K. A., van Dokkum, P. G., Wu, P.-F., Gallazzi, A., and Zibetti, S., MNRAS, 532, 3604, DOI: ([10.1093/mnras/stae1739](https://doi.org/10.1093/mnras/stae1739))
2. *The JWST-SUSPENSE Ultradeep Spectroscopic Program: Survey Overview and Star-Formation Histories of Quiescent Galaxies at $1 < z < 3$*
Slob, M., M. Kriek, **A. Beverage**, K. A. Suess, G. Barro, R. Bezanson, G. Brammer, C. M. Cheng, C. Conroy, A. de Graaff, N. M. Förster Schreiber, M. Franx, B. Lorenz, P. E. Mancera Piña, D. Marchesini, A. Muzzin, A. B. Newman, S. H. Price, A. E. Shapley, M. Stefanon, P. van Dokkum, and D. R. Weisz, 2024, ApJ, 973, 131, DOI: ([10.3847/1538-4357/ad65ff](https://doi.org/10.3847/1538-4357/ad65ff))
1. *The Heavy Metal Survey: Star Formation Constraints and Dynamical Masses of 21 Massive Quiescent Galaxies at $z = 1.3-2.3$*
Kriek, M., **A. Beverage**, S. H. Price, K. A. Suess, G. Barro, R. S. Bezanson, C. Conroy, S. E. Cutler, M. Franx, J. Lin, B. Lorenz, Y. Ma, I. G. Momcheva, L. A. Mowla, I. Pasha, P. van Dokkum, and K. E. Whitaker, 2024, ApJ, 966, 1, DOI: [10.3847/1538-4357/ad2df9](https://doi.org/10.3847/1538-4357/ad2df9)

SEMINARS

University of Minnesota	Colloquium, 2026
Texas A&M	Colloquium, 2026
University of Halifax	Astronomy Seminar, 2026
UCLA	Colloquium, 2026
Leiden	Colloquium, 2026
UC Davis	Cosmology Seminar, 2026
Carnegie Observatories	Lunch Seminar, 2024
Caltech Department of Astronomy	Seminar, 2024
Washington State University	Colloquium, 2023
Ohio State University	Galaxy Seminar, 2024
Leiden University	Dusty Galaxies Seminar, 2022
University of Edinburgh	Coffee Talk, 2023
Princeton University	Galread, 2022
UC Berkeley	Lunch Talk, 2021, 2023

CONFERENCE TALKS (*INVITED)

Tokyo, JP, The Universe in Full Bloom	2026
*UIUC, The End of Star Formation	2026
*Sesto, IT, Cosmic Nuggets II	2026
*Sesto, IT, The Growth of Galaxies in the Early Universe	2026
*Northwestern, CIERA Fellows at 15 Conference	2025
*Leiden, NL, Lorentz Workshop on Big Galaxies, Big Problems	2025
*Yale, The Inaugural Tinsley Workshop	2024
*UC Santa Cruz, Galaxy Workshop	2023
Sirolo, Italy, AGN Feedback and Star Formation Across Cosmic Scales and Time	2024
Cape Town, South Africa, IAU: The First Chapters of Our Cosmic History with <i>JWST</i>	2024
Tenerife, Spain, A Life Devoted to Stellar Populations	2023
Berkeley, USA, Keck Science Meeting	2023
Catania, Italy, Charting the Metallicity Evolution History of the Universe	2022
Baltimore, USA, STScI Multi-Object Spectroscopy Workshop	2021

TEACHING & ADVISING

Research mentor to Graduate Student Rana Osman, Princeton University	2026
Publication: <i>The chemical compositions of massive galaxies at $z \sim 2$ (in prep.)</i>	
Research mentor to Yilun Ma, University of California Berkeley	2020 – 2022
Publication: <i>Two Transitional Galaxies with AGN-Driven Outflows at $z \sim 2$ (in prep.)</i>	
* Graduate Student Instructor , UC Berkeley 7B: Introduction to Astrophysics II	Spring 2020
Graduate Student Instructor , UC Berkeley 7A: Introduction to Astrophysics I	Fall 2019
*Awarded Outstanding Astronomy GSI	

OUTREACH, SERVICE & DEI

Referee The Astrophysical Journal, Astronomy & Astrophysics	2021 – Present
POWER Coordinator	2022 – 2025
<i>POWER is a graduate student-led mentoring program and workshop series for Bay Area community college students interested in the physical sciences. I organized and led over 5 workshops that reached over 100 students and included financial aid and support network presentations, career and transfer student panels, and lab tours.</i>	
MPS Scholars Mentor	2023
<i>Professional development mentorship to a Berkeley undergrad</i>	
MPS Scholars Retreat	Spring 2024
<i>Graduate student panelist at professional development retreat for Berkeley undergrads</i>	
POWER Mentor	2021
<i>Professional development mentorship to a Bay Area community college student</i>	
Climate & DEI Committee Rep	2020 – 2021
<i>Graduate student representative on the Berkeley Astronomy DEI committee</i>	
Respect is Part of Research, Facilitator	2021
Grad Student Postdoc Seminar, Organizer	2022 – 2025